Endometriosis AI Detection Assistant

An AI-powered tool designed to close the diagnostic gap — using natural language processing to analyze symptom descriptions, generate explainable insights, and empower patients to seek care sooner.

The Problem

- Endometriosis affects ~10% of women of reproductive age, yet remains chronically underdiagnosed
- The average diagnostic journey spans **7–10 years**
- Symptoms are complex, overlapping, and frequently dismissed by clinicians

The Solution

- AI analyzes free-text symptom descriptions using NLP
- Generates structured, explainable clinical insights
- Encourages timely, informed medical consultation

🎯 Improve Awareness

Support early screening and symptom recognition across underserved populations

⚖️ Ethical AI

Demonstrate responsible, non-diagnostic AI design in a sensitive clinical domain

🔭 Scalable Vision

Built to expand toward multimodal AI — integrating text and medical imaging

ARCHITECTURE

System Architecture

The pipeline transforms raw patient-reported symptoms into structured, explainable outputs — designed for clarity, privacy, and responsible guidance.

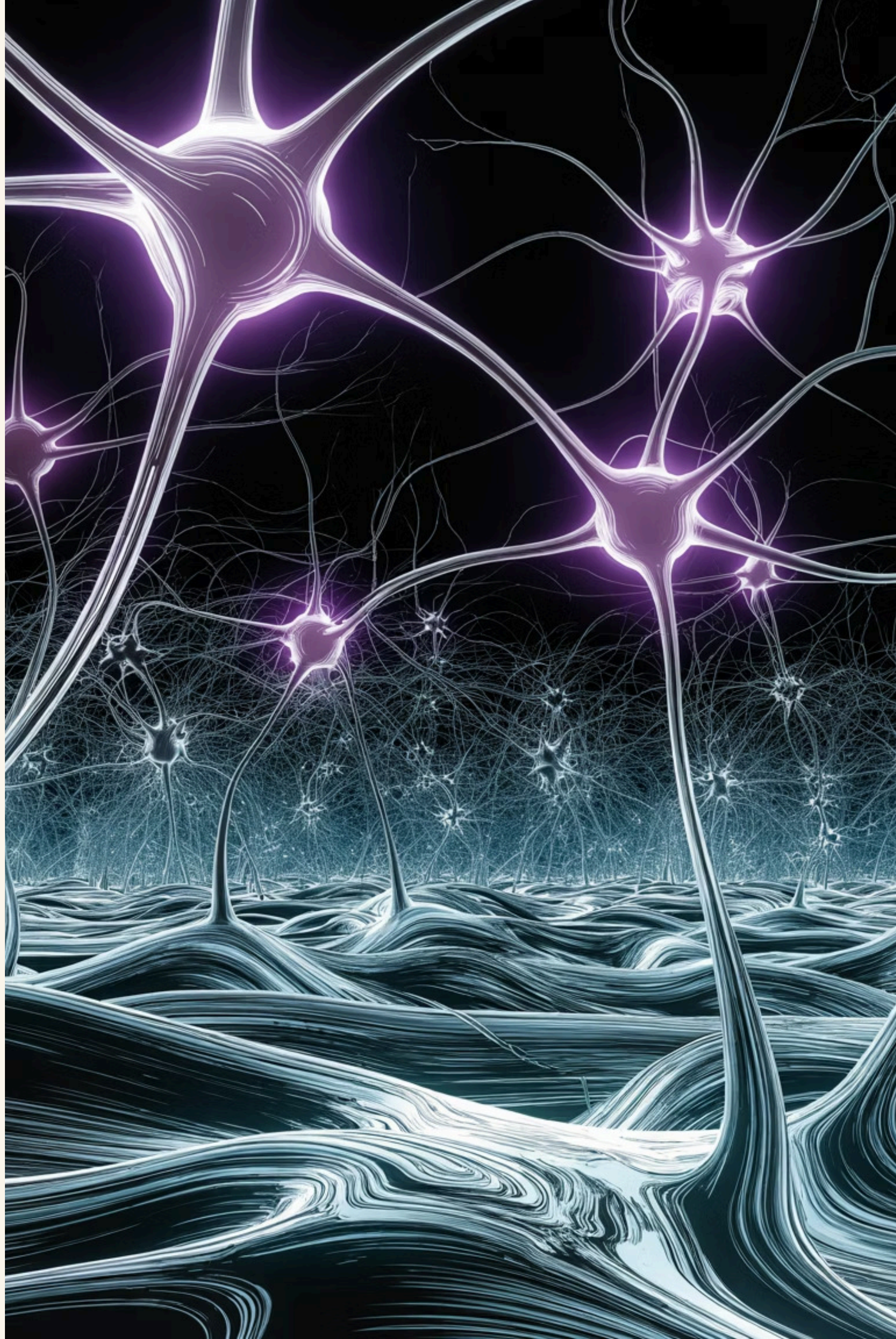
User Input

Text → NLP Model [Future Images → CV Model]

Combined Analysis

Output

At each stage, the system is designed to remain interpretable — so clinicians and patients alike can understand *why* a recommendation was made, not just what it says.



UNDER THE HOOD

Technology Stack & Design Principles

Core Technologies

01

Python

Core language for model development and data processing pipelines

02

Streamlit

Rapid, interactive front-end for clinical prototype demonstration

03

TensorFlow

Deep learning framework powering the NLP classification model

04

Pandas / NumPy

Data wrangling, feature engineering, and statistical validation

Design Philosophy

Explainable Outputs

Every confidence score is accompanied by a plain-language rationale — not a black box verdict

Privacy-First Processing

Symptom data is processed locally with no persistent storage of personally identifiable information

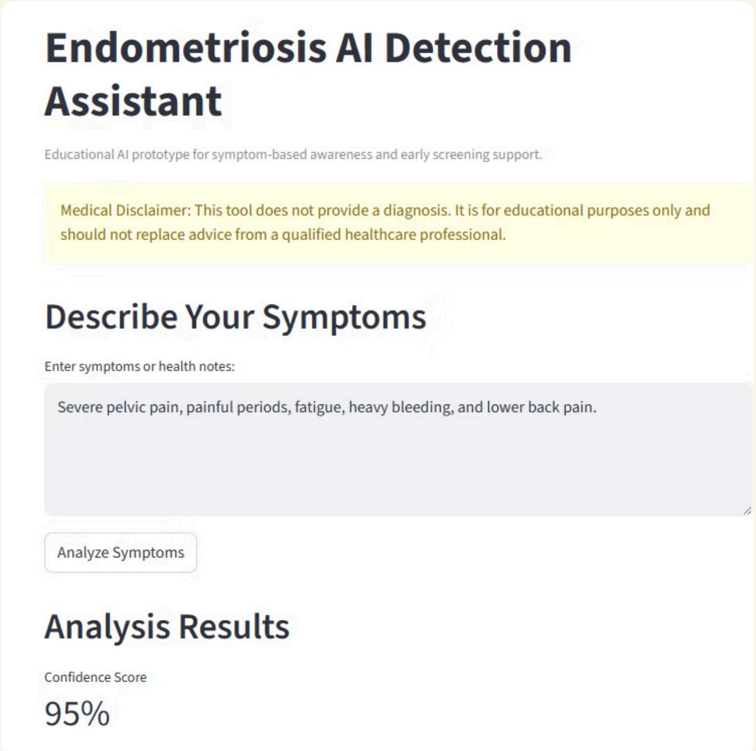
Non-Diagnostic Guidance

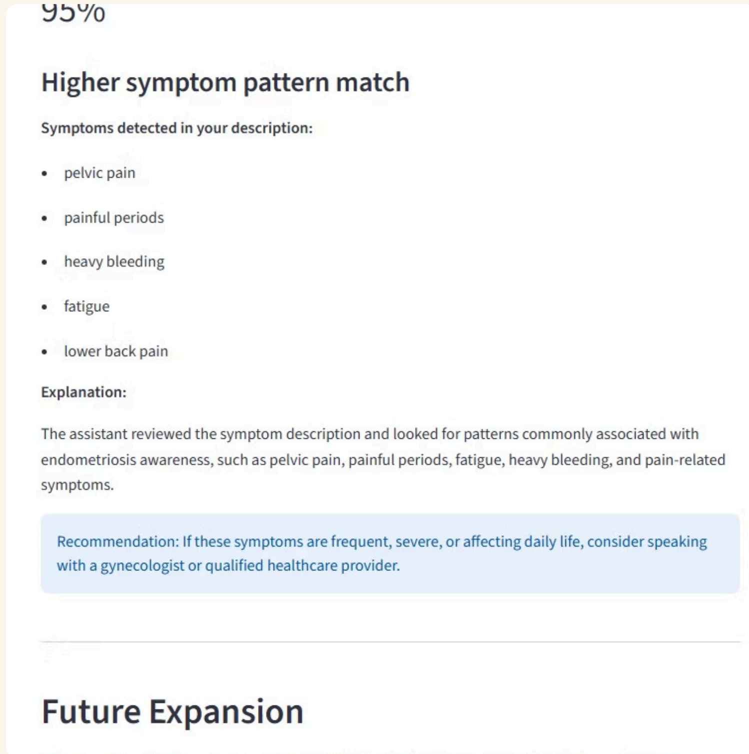
The tool is a screening aid, not a diagnostic instrument — always deferring to clinical authority

LIVE DEMO

Results & Demo

The prototype accepts natural language symptom descriptions and returns a structured, explainable analysis — demonstrating that responsible AI can deliver clinically meaningful guidance at scale.





Example Input

"*Severe pelvic pain, chronic fatigue, irregular menstrual cycles, bloating after meals*"

The model accepts free-text, just as a patient would naturally describe their experience — no structured forms required.

Example Output

- **Confidence Level:** 68% symptom-pattern alignment
- **Symptom Explanation:** The model identified patterns across pelvic pain, fatigue, and cycle-related symptoms.
- **Recommendation:** Consult a gynecologist or pelvic health specialist for further evaluation

Outcome: A fully functional AI prototype for symptom-based screening — demonstrating the power of NLP and responsible AI design in a real-world healthcare context.